



**Generation Interconnection
System Impact Study Report
for
Queue Project AF1-311
MONTOUR 230 KV
57 MW Capacity / 150 MW Energy**

August 2020

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is PPL.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 General

The Interconnection Customer (IC), **sPower Development Company, LLC**, has proposed a Solar generating facility located in Montour County, Pennsylvania. The installed facilities will have a total capability of 150 MW with 57 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is September 30, 2022. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-311
Project Name	MONTOUR 230 KV
State	Pennsylvania
County	Montour
Transmission Owner	PPL
MFO	150
MWE	150
MWC	57
Fuel	Solar
Basecase Study Year	2023

New Service Customers proposing queue projects that can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF1-311 will interconnect with the PPL on transmission system at the Montour 230 kV substation.

AF1-311 will interconnect with the PPL EU transmission system via new breaker-and-a-half bay position in Montour 230 kV substation. The Point of Interconnection (POI) will be at the PPL EU owned termination structure onto which the Interconnection Customer's transmission line terminates (with insulators) outside the Montour 230 kV substation.

5 Cost Summary

The AF1-311 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ 2,591,000
Allocation towards System Network Upgrade Costs*	\$ 14,315,071
Total Costs	\$ 16,906,071

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your project's present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your project's present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

The total physical interconnection costs is given in the table below:

Description	Total Cost
Attachment Facilities	\$ 325,000
Direct Connection Network Upgrade	\$ 0
Non Direct Connection Network Upgrades	\$ 2,266,000
Total Physical Interconnection Costs	\$ 2,591,000

PPL EU will populate existing Montour 230 kV Bay 6 with two circuit breakers and associated equipment to complete a full breaker-and-a-half bay to accommodate a new line position for AF1-311.

Risks and Assumptions

- No major environmental, real estate, siting, geotechnical, or permitting issues
- IC is responsible for acquisition of easements, permits, and right of way for any Direct Connection Network Upgrades and Attachment Facilities per PPL EU standards and requirements.

- PPL EU will perform all grading, site preparation, and establish access roads for the PPL EU owned Attachment Facilities per PPL EU standards and requirements.
- Pennsylvania Public Utility Commission will approve the PPL EU-prepared/submitted Letter of Notification for the PPL EU owned Attachment Facilities

6.1 Attachment Facilities

230 kV Transmission Line Tap

PPL EU will construct the Attachment Facilities from the dead-end structure inside the Montour Substation to a PPL EU owned termination structure (POI structure) outside the Montour Substation. The IC will be responsible for constructing the Attachment Facilities from the POI structure to their collector substation.

PPL EU and the IC will need to consider the required spacing and easements for AE2-271 and AF2-331 which are connecting adjacent to AF1-311 in Bays 5 and 7, respectively. Each request will have a POI as an H-frame dead-end structure outside the Montour Substation.

For the purposes of this report, PPL EU is assuming the conductor spacing and clearances is sufficient to accommodate AF1-311. If during detailed design, spacing is determined not to be adequate, PPL EU will need to design an underground solution and/or relocate existing PPL EU assets. This evaluation will occur during the Facilities Study.

PPL EU will perform the following work for the Attachment Facilities:

- Install one (1) H-frame terminal tension structure with custom foundation and fiber splice box outside the Montour Substation
- Route 48-ct fiber ADSS from the splice box on the above structure underground to the control cubicle inside the Montour Substation
- 260 ft. of 1590 ACSR conductor from the substation dead-end structure to the H-frame structure outside the Montour Substation
- Submit a Letter of Notification to, and receive approval from, the Pennsylvania Public Utility Commission for siting of the PPL EU owned Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
230 kV Transmission Line Tap	\$ 325,000
Total Attachment Facility Costs	\$ 325,000

6.2 Direct Connection Cost Estimate

None.

6.3 Non-Direct Connection Cost Estimate

Montour 230 kV Substation- Populate Existing Bay 6

- Populate the existing Bay 6 by adding Two (2) new circuit breakers and associated equipment to complete a full breaker and a half bay. The Attachment Facilities will terminate in Bay 6 at the newly created line position dead-end structure.
- Route 48-count fiber terminated on the new A-frame structure in the substation and route it into the control cubicle

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Montour Substation – Populate Existing Bay 6	\$ 2,266,000
Total Non-Direct Connection Facility Costs	\$ 2,266,000

7 Schedule

The estimated time to complete the scope of work is **18 - 24 months** after the PJM three-party Interconnection Service Agreement (ISA) and Interconnection Construction Service Agreement (ICSA) are signed and PPL EU receives Notice to Proceed from the IC.

8 Interconnection Customer Requirements

8.1 PPL EU Interconnection Requirements

PPL EU applicable technical standards that address requirements for interconnection of generation, transmission, and end user facilities can be found at the following link:

<https://pjm.com/planning/design-engineering/to-tech-standards/private-ppl.aspx>

9 Revenue Metering and SCADA Requirements

9.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

9.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

9.3 Interconnected Transmission Owner Requirements

Installation of revenue grade Bi-directional Metering Equipment will be required in the vicinity of the POI to measure kWh and kVARh. PPL EU will design and supply the required metering equipment; all installation costs would be borne by the IC including CTs/PTs. All metering equipment must meet applicable PPL EU tariff requirements as well as being compliant with all applicable requirements of the PJM agreements. The equipment must provide bidirectional revenue metering (kWh and kVARh) and real-time data (kW, kVAR, circuit breaker status, and generator bus voltages) for the IC's generating resource. The metering equipment should be housed in a control cabinet or similar enclosure and must be accessible to PPL EU metering personnel.

10 Summer Peak Analysis

The Queue Project AF1-311 was evaluated as a 150.0 MW (Capacity 57.0 MW) injection at the **Montour 230 kV substation** in the PPL area. Project AF1-311 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-311 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

10.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

10.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS#	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42311967	208034	MILT	230	PPL	208109	SUNB	230	PPL	1	PL_P42_000488	breaker	730.0	95.77	102.21	AC	47.73
42311968	208034	MILT	230	PPL	208109	SUNB	230	PPL	1	PL_P42_100989	breaker	730.0	95.77	102.21	AC	47.73
42816639	208034	MILT	230	PPL	208109	SUNB	230	PPL	1	PL_P71_001414	tower	730.0	96.09	101.47	AC	40.29
42816612	208040	MONT	230	PPL	208034	MILT	230	PPL	1	PL_P71_101343	tower	801.0	96.5	101.45	AC	40.29

10.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS#	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42816638	208034	MILT	230	PPL	208109	SUNB	230	PPL	1	PL_P71_101343	tower	730	102.55	107.95	AC	40.29
42311788	938390	AE1-058 TAP	230	PPL	208072	SIEG	230	PPL	1	PJM500_PL_P42_000922	breaker	628	107.58	109.48	AC	13.9
42311789	938390	AE1-058 TAP	230	PPL	208072	SIEG	230	PPL	1	PJM500_PL_P42_000923	breaker	628	107.58	109.48	AC	13.9

10.4 Steady-State Voltage Requirements

To be determined

10.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/DC	MW IMPACT
42631886	208034	MILT	230	PPL	208109	SUNB	230	PPL	1	PL_P1 3_100 455	operation	730	94.15	100.32	AC	45.63
42631887	208034	MILT	230	PPL	208109	SUNB	230	PPL	1	PL_P1 2_000 218	operation	730	94.15	100.32	AC	45.63
42631608	938390	AE1-058 TAP	230	PPL	208072	SIEG	230	PPL	1	PJM50 0_PL_P 12_00 0083	operation	628	107.58	109.39	AC	13.32

10.6 System Reinforcements

ID	Index	Facility	Upgrade Description	Upgrade Cost	AF1-311 Cost Allocation																				
42311967, 42311968, 42816639, 42816612, 42816638	1, 2	MILT 230.0kV – SUNB 230.0kV Ckt 1 MONT 230.0kV – MILT 230.0kV Ckt 1	Project ID : S1106 Description : PPL Supplemental project to rebuild the Montour - Milton 230kV line to double circuit and change operating voltage 69kV line between Milton and Sunbury to 230kV Type : CONTINGENCY Total Cost : \$63,000,000 Time Estimate : 36 Months	\$63,000,000	\$0																				
42311788, 42311789	3	AE1-058 TAP 230.0kV - SIEG 230.0kV Ckt 1	Project ID : N6406 (R-PL-0003A) Description : Rebuild AE1-058 yard-SIEG 230kV line Type : FACILITY Total Cost : \$64,700,000 Time Estimate : 71 Months Ratings : 665/821/821 <table><tr><th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th></tr><tr><td>AE2-110</td><td>1.31</td><td>2.09%</td><td>\$1,353,238</td></tr><tr><td>AE2-271</td><td>9.51</td><td>15.14%</td><td>\$9,793,980</td></tr><tr><td>AE2-295</td><td>38.1</td><td>60.65%</td><td>\$39,237,712</td></tr><tr><td>AF1-311</td><td>13.9</td><td>22.13%</td><td>\$14,315,071</td></tr></table>	Queue	MW	Cost %	Cost \$	AE2-110	1.31	2.09%	\$1,353,238	AE2-271	9.51	15.14%	\$9,793,980	AE2-295	38.1	60.65%	\$39,237,712	AF1-311	13.9	22.13%	\$14,315,071	\$64,700,000	\$14,315,071
Queue	MW	Cost %	Cost \$																						
AE2-110	1.31	2.09%	\$1,353,238																						
AE2-271	9.51	15.14%	\$9,793,980																						
AE2-295	38.1	60.65%	\$39,237,712																						
AF1-311	13.9	22.13%	\$14,315,071																						
			Total Cost	\$127,700,000	\$14,315,071																				

Note: For customers with System Reinforcements listed: If your project's present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your project's present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

10.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

10.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42816638	208034	MILT	PPL	208109	SUNB	PPL	1	PL_P71_101343	tower	730.0	102.55	107.95	AC	40.29

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
208911	MONT G1	36.0797	50/50	36.0797
208912	MONT G2 (Deactivation : 18/02/2019)	36.5459	50/50	36.5459
209006	NEPC IPP (Deactivation : 24/10/2018)	1.6520	Adder	1.94
211369	W1-111 BAT	0.0064	Adder	0.01
211375	BEAC	1.2707	Adder	1.49
211418	BUMO	0.8662	Adder	1.02
212369	PATRIOT 1	18.1917	50/50	18.1917
212370	PATRIOT 2	18.1917	50/50	18.1917
292935	U2-015E OP1	5.7969	Adder	6.82
921653	AA2-008 E	15.3062	50/50	15.3062
923673	AB1-182 E	1.2128	Adder	1.43
938401	AE1-059 C O1	10.0895	Adder	11.87
939521	AE1-181 C	1.7155	Adder	2.02
939522	AE1-181 E	1.1437	Adder	1.35
940561	AE2-042 C O1	27.4033	50/50	27.4033
940562	AE2-042 E O1	13.5845	50/50	13.5845
940592	AE2-046 E	1.6520	Adder	1.94
940721	AE2-059 C	4.9185	50/50	4.9185
940722	AE2-059 E	6.7923	50/50	6.7923
940941	AE2-084 C	4.9185	50/50	4.9185
940942	AE2-084 E	6.7923	50/50	6.7923
941161	AE2-110 C	1.2794	50/50	1.2794
941162	AE2-110 E	1.7668	50/50	1.7668
942281	AE2-241 C	1.2794	50/50	1.2794
942282	AE2-241 E	1.7668	50/50	1.7668
942561	AE2-271 C O1	16.3828	50/50	16.3828
942562	AE2-271 E O1	10.9039	50/50	10.9039
943723	AF1-040 BAT	2.4922	50/50	2.4922
945611	AF1-226 C	2.2390	50/50	2.2390
945612	AF1-226 E	3.0919	50/50	3.0919
946471	AF1-311 C O1	15.3085	50/50	15.3085
946472	AF1-311 E O1	24.9770	50/50	24.9770
NEWTON	NEWTON	0.8263	Confirmed LTF	0.8263
FARMERCITY	FARMERCITY	0.0429	Confirmed LTF	0.0429
G-007A	G-007A	2.2728	Confirmed LTF	2.2728
VFT	VFT	7.0950	Confirmed LTF	7.0950
PRAIRIE	PRAIRIE	1.9760	Confirmed LTF	1.9760
COFFEEN	COFFEEN	0.1536	Confirmed LTF	0.1536
CHEOAH	CHEOAH	0.3694	Confirmed LTF	0.3694
EDWARDS	EDWARDS	0.2709	Confirmed LTF	0.2709
TILTON	TILTON	0.4870	Confirmed LTF	0.4870

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
MADISON	MADISON	0.0141	Confirmed LTF	0.0141
GIBSON	GIBSON	0.4204	Confirmed LTF	0.4204
CALDERWOOD	CALDERWOOD	0.3673	Confirmed LTF	0.3673
BLUEG	BLUEG	1.3367	Confirmed LTF	1.3367
TRIMBLE	TRIMBLE	0.4285	Confirmed LTF	0.4285
CATAWBA	CATAWBA	0.2495	Confirmed LTF	0.2495

10.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42816612	208040	MONT	PPL	208034	MILT	PPL	1	PL_P71_101343	tower	801.0	96.5	101.45	AC	40.29

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
208911	MONT G1	36.0797	50/50	36.0797
208912	MONT G2 (Deactivation : 18/02/2019)	36.5459	50/50	36.5459
209006	NEPC IPP (Deactivation : 24/10/2018)	1.6520	Adder	1.94
211369	W1-111 BAT	0.0064	Adder	0.01
211375	BEAC	1.2707	Adder	1.49
211418	BUMO	0.8662	Adder	1.02
212369	PATRIOT 1	18.1917	50/50	18.1917
212370	PATRIOT 2	18.1917	50/50	18.1917
292935	U2-015E OP1	5.7969	Adder	6.82
921653	AA2-008 E	15.3062	50/50	15.3062
923673	AB1-182 E	1.2128	Adder	1.43
938401	AE1-059 C O1	10.0895	Adder	11.87
939521	AE1-181 C	1.7155	Adder	2.02
939522	AE1-181 E	1.1437	Adder	1.35
940592	AE2-046 E	1.6520	Adder	1.94
941161	AE2-110 C	1.2794	50/50	1.2794
941162	AE2-110 E	1.7668	50/50	1.7668
942281	AE2-241 C	1.2794	50/50	1.2794
942282	AE2-241 E	1.7668	50/50	1.7668
942561	AE2-271 C O1	16.3828	50/50	16.3828
942562	AE2-271 E O1	10.9039	50/50	10.9039
943723	AF1-040 BAT	2.4922	50/50	2.4922
945611	AF1-226 C	2.2390	50/50	2.2390
945612	AF1-226 E	3.0919	50/50	3.0919
946471	AF1-311 C O1	15.3085	50/50	15.3085
946472	AF1-311 E O1	24.9770	50/50	24.9770
NEWTON	NEWTON	0.8263	Confirmed LTF	0.8263
FARMERCITY	FARMERCITY	0.0429	Confirmed LTF	0.0429
G-007A	G-007A	2.2728	Confirmed LTF	2.2728
VFT	VFT	7.0950	Confirmed LTF	7.0950
PRAIRIE	PRAIRIE	1.9760	Confirmed LTF	1.9760
COFFEEN	COFFEEN	0.1536	Confirmed LTF	0.1536

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CHEOAH	CHEOAH	0.3694	Confirmed LTF	0.3694
EDWARDS	EDWARDS	0.2709	Confirmed LTF	0.2709
TILTON	TILTON	0.4870	Confirmed LTF	0.4870
MADISON	MADISON	0.0141	Confirmed LTF	0.0141
GIBSON	GIBSON	0.4204	Confirmed LTF	0.4204
CALDERWOOD	CALDERWOOD	0.3673	Confirmed LTF	0.3673
BLUEG	BLUEG	1.3367	Confirmed LTF	1.3367
TRIMBLE	TRIMBLE	0.4285	Confirmed LTF	0.4285
CATAWBA	CATAWBA	0.2495	Confirmed LTF	0.2495

10.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42311789	938390	AE1-058 TAP	PPL	208072	SIEG	PPL	1	PJM500_PL_P42_000923	breaker	628.0	107.58	109.48	AC	13.9

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
208941	FISH CT	1.3543	50/50	1.3543
208981	FOWH IPP	1.6701	50/50	1.6701
208982	GLBT IPP	3.9663	50/50	3.9663
209013	SCEN IPP	4.1598	50/50	4.1598
209018	SUNBIPCT	0.6748	50/50	0.6748
209019	VIKI IPP	1.4394	Adder	1.69
209021	WEST IPP	1.2711	50/50	1.2711
209022	WHFR IPP (Deactivation : 03/03/2020 WD - 01/03/2020)	11.7433	50/50	11.7433
209027	LOR2_Q27	0.9674	50/50	0.9674
212099	BRMO IPP	0.2699	50/50	0.2699
212174	INGE	0.2386	50/50	0.2386
212266	LOR1	0.2660	50/50	0.2660
920651	AA2-171 E	2.4531	Adder	2.89
920711	AA2-182 C (Withdrawn : 05/04/2020)	64.7751	Adder	76.21
921653	AA2-008 E	4.5843	Adder	5.39
924291	AB2-074 C (Withdrawn : 05/04/2020)	3.3150	Adder	3.9
924292	AB2-074 E (Withdrawn : 05/04/2020)	4.1769	Adder	4.91
935083	AD1-143 BAT1 (Suspended)	0.4957	Merchant Transmission	0.4957
935103	AD1-143 BAT2 (Suspended)	0.4957	Merchant Transmission	0.4957
938391	AE1-058 C	88.3675	50/50	88.3675
938392	AE1-058 E	88.3675	50/50	88.3675
939891	AE1-225 C O1	0.8457	Adder	0.99
939892	AE1-225 E O1	0.9356	Adder	1.1

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940561	AE2-042 C O1	4.1184	Adder	4.85
940562	AE2-042 E O1	2.0416	Adder	2.4
940721	AE2-059 C	0.7392	Adder	0.87
940722	AE2-059 E	1.0208	Adder	1.2
940941	AE2-084 C	0.7392	Adder	0.87
940942	AE2-084 E	1.0208	Adder	1.2
941161	AE2-110 C	0.9210	50/50	0.9210
941162	AE2-110 E	1.2718	50/50	1.2718
941171	AE2-111 C	0.7246	Adder	0.85
941172	AE2-111 E	1.0007	Adder	1.18
941371	AE2-133 C	0.6940	Adder	0.82
941372	AE2-133 E	0.9584	Adder	1.13
942281	AE2-241 C	0.9210	50/50	0.9210
942282	AE2-241 E	1.2718	50/50	1.2718
942561	AE2-271 C O1	4.8044	Adder	5.65
942562	AE2-271 E O1	3.1977	Adder	3.76
942771	AE2-295 C O1	5.6576	50/50	5.6576
942772	AE2-295 E O1	32.8229	50/50	32.8229
943721	AF1-040 C	0.0900	Adder	0.11
943722	AF1-040 E	2.0110	50/50	2.0110
945511	AF1-216 C1O1	3.0977	Adder	3.64
945512	AF1-216 E1O1	2.0627	Adder	2.43
945521	AF1-217 C O2	0.6134	Adder	0.72
945522	AF1-217 E O2	0.4090	Adder	0.48
945611	AF1-226 C	1.6117	50/50	1.6117
945612	AF1-226 E	2.2257	50/50	2.2257
946221	AF1-287 C	0.6080	Adder	0.72
946222	AF1-287 E	0.4053	Adder	0.48
946471	AF1-311 C O1	4.4894	Adder	5.28
946472	AF1-311 E O1	7.3248	Adder	8.62
946691	AF1-333 C O1	0.9851	Adder	1.16
946692	AF1-333 E O1	0.6567	Adder	0.77
946731	AF1-337 C	0.9851	Adder	1.16
946732	AF1-337 E	0.6567	Adder	0.77
946741	AF1-338 C	0.9851	Adder	1.16
946742	AF1-338 E	0.6567	Adder	0.77
946751	AF1-339 C O1	0.9851	Adder	1.16
946752	AF1-339 E O1	0.6567	Adder	0.77
946761	AF1-271A C	0.9176	Adder	1.08
946762	AF1-271A E	0.6118	Adder	0.72
NEWTON	NEWTON	0.0870	Confirmed LTF	0.0870
FARMERCITY	FARMERCITY	0.0046	Confirmed LTF	0.0046
NY	NY	0.0243	Confirmed LTF	0.0243
PRAIRIE	PRAIRIE	0.2196	Confirmed LTF	0.2196
O-066	O-066	7.5533	Confirmed LTF	7.5533
COFFEEN	COFFEEN	0.0160	Confirmed LTF	0.0160
CHEOAH	CHEOAH	0.0541	Confirmed LTF	0.0541
EDWARDS	EDWARDS	0.0266	Confirmed LTF	0.0266
TILTON	TILTON	0.0485	Confirmed LTF	0.0485
G-007	G-007	0.9495	Confirmed LTF	0.9495
GIBSON	GIBSON	0.0437	Confirmed LTF	0.0437
CALDERWOOD	CALDERWOOD	0.0532	Confirmed LTF	0.0532

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
BLUEG	BLUEG	0.1406	Confirmed LTF	0.1406
TRIMBLE	TRIMBLE	0.0445	Confirmed LTF	0.0445
CATAWBA	CATAWBA	0.0452	Confirmed LTF	0.0452

10.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA2-008	Saegers 230 kV	In Service
AA2-171	Sunbury #1 500kV	In Service
AA2-182	Sunbury #2 500kV	Withdrawn
AB1-182	Bear Creek	Engineering and Procurement
AB2-074	Sunbury #2 500kV	Withdrawn
AD1-143	Hauto-Siegfried 69 kV	Suspended
AE1-058	Frackville-Siegfried 230 kV	Active
AE1-059	Stanton-Summit 230 kV	Active
AE1-181	St. Johns-Freeland 69 kV	Active
AE1-225	Columbia-Sunbury 69 kV	Active
AE2-042	Milton 69 kV	Active
AE2-046	Harwood-East Hazelton 69 kV	Active
AE2-059	Derry Tap-Derry Bus 69 kV	Active
AE2-084	Derry Tap-Derry Bus 69 kV	Active
AE2-110	Columbia-West Bloomsburg 69 kV	Active
AE2-111	Beavertown Tap-Beavertown Weaving Tap 69kV	Active
AE2-133	Penns Tap-Richfield Tie 69 kV	Active
AE2-241	Bloomsburg-Columbia 69 kV	Active
AE2-271	Montour 230 kV	Active
AE2-295	Eldred 230 kV	Active
AF1-040	Dauphin-Pine Grove 69 kV	Active
AF1-216	Lycoming-Lock Haven 69 kV	Active
AF1-217	Edinboro -Cambridge Springs 34.5 kV	Active
AF1-226	Bowmans Mill-Scott 69 kV	Active
AF1-271A	Gratz 69 kV	Active
AF1-287	Edinboro South 34.5 kV II	Active
AF1-311	Montour 230 kV	Active
AF1-333	Laurelton-Mifflinburg 69 kV	Active
AF1-337	Laurelton-Mifflinburg 69 kV	Active
AF1-338	Laurelton-Mifflinburg 69 kV	Active
AF1-339	Laurelton-Mifflinburg 69 kV	Active
U2-015	Harwood-E. Palmerton 230kV	Withdrawn
W1-111	Harwood-Berwick 69kV	In Service

10.9 Contingency Descriptions

Contingency Name	Contingency Definition
PL_P42_100989	CONTINGENCY 'PL_P42_100989' /* MONT 230 4T BF COLUMBIA & SAEGERS TIE CB DISCONNECT BUS 207943 /* /* COLU-MONT & COLU T2 DISCONNECT BRANCH FROM BUS 208040 TO BUS 212397 CKT 1 /* /* MONT-SAEG 230 PROJECT B1602 END
PL_P13_100455	CONTINGENCY 'PL_P13_100455' /* COLUMBIA 230/69KV TRAN 2 OUT DISCONNECT BUS 207943 /* END
PL_P42_000488	CONTINGENCY 'PL_P42_000488' /* MONT-COLU 230 KV STUCK BREAKER CONNECTED TO MONT-SAEG2 LINE DISCONNECT BRANCH FROM BUS 208040 TO BUS 212397 CKT 2 /* MONT-SAEG 230 DISCONNECT BRANCH FROM BUS 207943 TO BUS 208040 CKT 1 /* COLU TR2-MONT 230 DISCONNECT BRANCH FROM BUS 207943 TO BUS 212093 CKT 2 /* COLU TR2-COLU 230-69 END
PJM500_PL_P12_000083	CONTINGENCY 'PJM500_PL_P12_000083' /* JUNI-SUNB 500KV LINE DISCONNECT BRANCH FROM BUS 200009 TO BUS 200021 CKT 1 /* JUNIATA-SUNBURY 500 END
PL_P71_101343	CONTINGENCY 'PL_P71_101343' /* SAEGERS-CLINTON & SAEGERS-ELIMSPORT 230 KV TOWER DISCONNECT BRANCH FROM BUS 212397 TO BUS 207968 CKT 1 /* /* SAEG-ELIM DISCONNECT BRANCH FROM BUS 207937 TO BUS 212397 CKT 1 /* /* SAEG-CLIN END
PL_P71_001414	CONTINGENCY 'PL_P71_001414' /* SAEG-ELIM 230KV LINE & CLIN-ELIM 230KV LINE DISCONNECT BRANCH FROM BUS 212397 TO BUS 207968 CKT 1 /* /* SAEG-ELIM 230KV LINE DISCONNECT BRANCH FROM BUS 207937 TO BUS 207968 CKT 1 /* /* CLIN-ELIM 230KV LINE END
PL_P12_000218	CONTINGENCY 'PL_P12_000218' /* MONT-COLU 230 KV LINE DISCONNECT BRANCH FROM BUS 207943 TO BUS 208040 CKT 1 /* COLU TR2-MONT 230 DISCONNECT BRANCH FROM BUS 207943 TO BUS 212093 CKT 2 /* COLU TR2-COLU 230-69 END
PJM500_PL_P42_000922	CONTINGENCY 'PJM500_PL_P42_000922' /* SUNBURY 500KV YARD 3N BF DISCONNECT BRANCH FROM BUS 200021 TO BUS 208109 CKT 24 /* /* T24 DISCONNECT BRANCH FROM BUS 200021 TO BUS 200009 CKT 1 /* /* JUNIATA-SUNBURY 500KV LINE END
PJM500_PL_P42_000923	CONTINGENCY 'PJM500_PL_P42_000923' /* SUNBURY 500KV YARD 3T BF DISCONNECT BRANCH FROM BUS 200021 TO BUS 208109 CKT 25 /* /* T25 DISCONNECT BRANCH FROM BUS 200021 TO BUS 200009 CKT 1 /* /* JUNIATA-SUNBURY 500KV LINE END

11 Light Load Analysis

Light load analysis is not required for AF1-311.

12 Short Circuit Analysis

The following Breakers are overdutied:

None

13 Stability and Reactive Power

(Summary of the VAR requirements based upon the results of the dynamic studies)

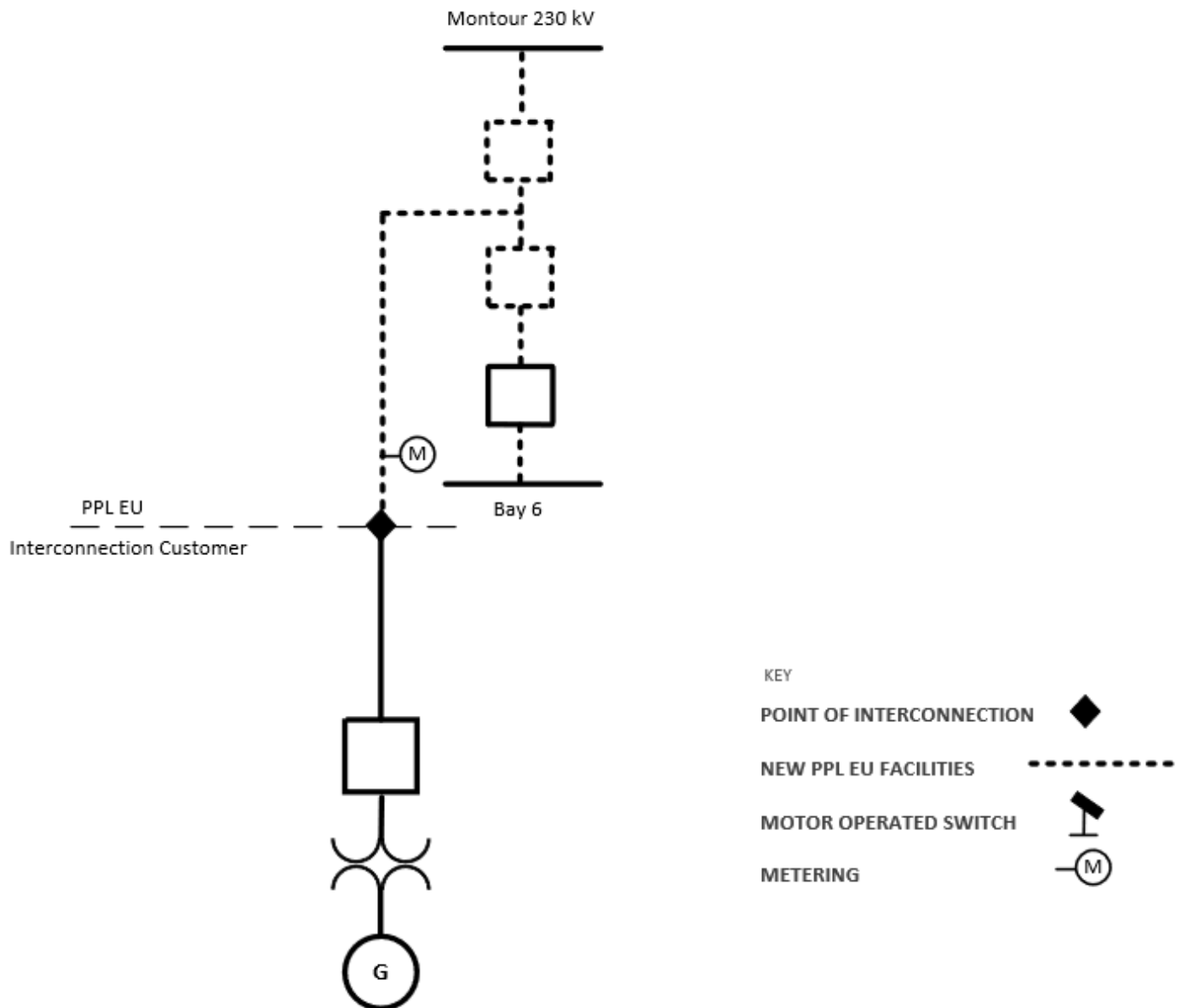
To be determined in the Facilities Study Phase. Additional reinforcements may be identified to mitigate stability concerns.

14 Affected Systems

14.1 NYISO

NYISO Impacts to be determined during later study phases (as applicable).

15 Attachment 1: One Line Diagram



16 Attachment 2: Site Location

